

HomeWork

Factory Design Pattern

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What's a design pattern?

Design patterns are typical solutions to commonly occurring problems in software design. They are like pre-made blueprints that you can customize to solve a recurring design problem in your code.



Types of Design Patterns



Creational



Structural



Behavioral

Problem

**monkey get into
twitter argument
(he's sending multiple
slurs)**



Design Pattern

- A design pattern provides a general reusable solution for the common problems that occur in software design.
- The pattern typically shows relationships and interactions between classes or objects.
- The idea is to speed up the development process by providing well-tested, proven development/design paradigms.

ain't no way



Intent of Factory Method

Factory Method is a creational design pattern that provides an interface for creating objects in a superclass but allows subclasses to alter the type of objects



Solution



When to use it?



Use the Factory Method when you want to provide users of your library or framework with a way to extend its internal components.



Use the Factory Method when you want to save system resources by reusing existing objects instead of rebuilding them each time

Pros and Cons



- You avoid tight coupling between the creator and the concrete products.
- *Single Responsibility Principle*. You can move the product creation code into one place in the program, making the code easier to support.
- *Open/Closed Principle*. You can introduce new types of products into the program without breaking existing client code.



- The code may become more complicated since you need to introduce a lot of new subclasses to implement the pattern. The best case scenario is when you're introducing the pattern into an existing hierarchy of creator classes.

Good Idea

It's not mandatory to always implement design patterns in your project. Design patterns are not meant for project development.

Design patterns are meant for common problem-solving. Whenever there is a need, you have to implement a suitable pattern to avoid such problems in the future.

To find out which pattern to use, you just have to try to understand the design patterns and their purposes. Only by doing that, you will be able to pick the right one.



**Trust the
process!**

